MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 6962

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: ATR, L-Band, Fo = 1,285 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	Du	Alt
Unit:	Mw	---	ft
Maximum:	2	0.001	10,000
Minimum:	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figures 1 and 2
Mounting:	See figure 1 and note 1
Weight:	6 pounds (approx)

TEST CONDITIONS:

Parameter: / Unit:	Incident power kw	tp μs	Du ---	VSWR σ'	F MHz
Test condition 1:					
Maximum:	525	4.4	0.001	1.05	1,298
Minimum:	475	3.6	---	---	1,272
Test condition 2:					
Maximum:	2,100	4.4	0.001	---	1,298
Minimum:	1,900	3.6	---	---	1,272

GENERAL:

Preproduction sample approval - Required (See note 2)

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	First article inspection (Preproduction)	2	---					
4461	Loaded Q	3	---		QL	---	2.5	---
4474	High-level VSWR	3	1		VSWR	---	1.15	---
4021	Degradation due to vibration	---	---		---	---	---	---
	Quality conformance inspection, part 1	4	---					
4482	Tuning susceptance	3	---	Method A	b	-0.03	+0.03	---
4486	Firing time	3	1		t	---	5	sec
4488	Arc loss	3	1		La	---	0.3	dB
1027	Temperature cycling	---	---	1 cycle	---	---	---	---
1027	Temperature cycling life test	---	---	Group D; 10 cycles (min)	---	---	---	---
	Quality conformance inspection, part 2							
4101	Dielectric material strain	---	---		---	---	---	---
4484	Normalized conductance	3	---		g	---	0.03	---
1031	Variable-frequency vibration	---	---	F = 50 to 500 Hz; 15 G; t = 30 minutes (min)	---	---	---	---
---	Insertion	5	---		---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	6	2	Group D	t	1,000	---	hrs
---	Life-test end points:	---	---					
4482	Tuning susceptance	3	---	Method A	b	-0.05	+0.05	---
4484	Normalized conductance	3	---		g	---	0.05	---
4488	Arc loss	3	1		La	---	0.4	dB

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NOTES:

1. Suitable RF gaskets will be supplied with each tube.

2. The number of First Article (Preproduction) samples required for inspection and the acceptance criteria shall be as specified for qualification testing in the basic document. All tests listed on the TSS are required for First Article (Preproduction) inspection; however, those tests grouped under this heading are to be performed during preproduction testing only.

3. This test shall be performed in a half-height L-band waveguide (ID6.500 x 1.625).

4. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

5. The tube shall be capable of being inserted and removed from its seat a minimum of 15 times with no deterioration in the tubes electrical characteristics.

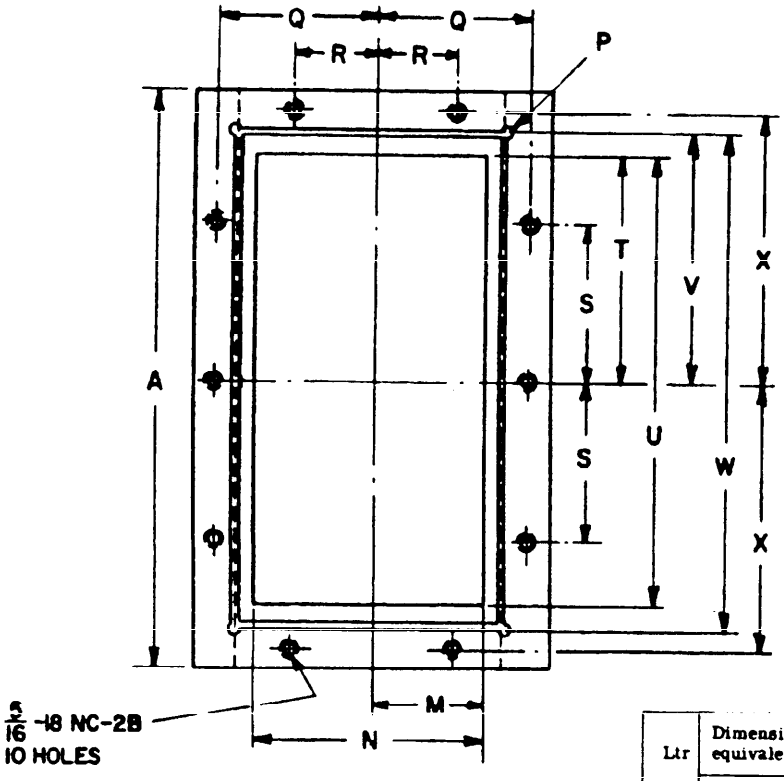
6. This test shall be performed in a full-height L-band waveguide (ID 6.500 x 3.250).

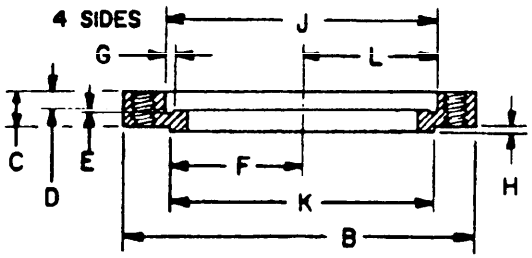
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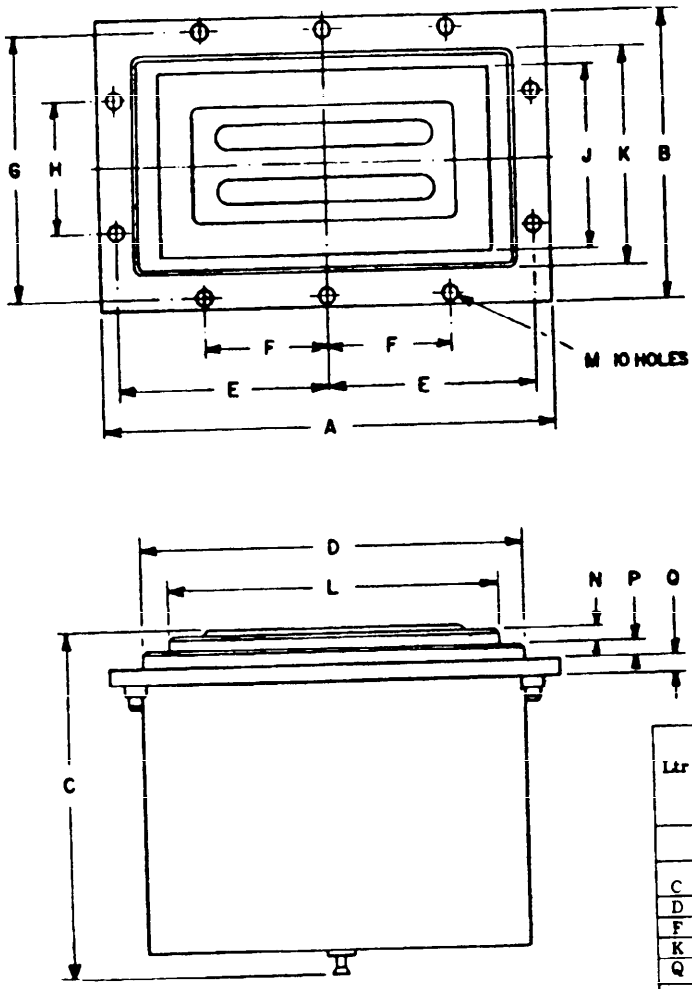




Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
Q	2.309 (58.65)	2.317 (58.85)
R	1.246 (31.65)	1.254 (31.85)
S	2.371 (60.22)	2.379 (60.43)
X	3.934 (99.92)	3.942(100.13)
Reference dimensions		
A	8.69 (220.73)	
B	5.437(138.10)	
C	.600 (15.24)	
D	.250 (6.35)	
E	.020 (.51)	
F	2.000 (50.80)	
G	.22 (5.59)	
H	.080 (2.03)	
J	4.135(105.03)	
K	4.000(101.60)	
L	2.067 (52.50)	
M	1.768 (44.91)	
N	3.537 (89.84)	
P	.22 x .270(5.59 x 6.86)DP	
T	3.393 (86.18)	
U	6.787(172.39)	
V	3.692 (93.78)	
W	7.385(187.58)	

FIGURE 1. Mounting seat.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Preproduction inspection		
C		6.125(155.58)
D	7.355(186.82)	7.375(187.33)
F	2.367 (60.12)	2.383 (60.53)
K	4.105(104.27)	4.125(104.78)
Q	.240 (6.10)	.260 (6.60)
Quality conformance inspection, part 1 (note 4)		
E	3.930 (99.82)	3.946(100.23)
J		3.475 (88.27)
L		6.725(170.82)
Quality conformance inspection, part 2		
G	4.618(117.30)	4.634(117.70)
H	2.492 (63.30)	2.508 (63.70)
M	.348(8.84) DRILL	
N	.370 (9.40)	.390 (9.91)
P	.240 (6.10)	.260 (6.60)
Reference dimensions		
A	8.69(220.73)	
B	5.44(138.18)	

FIGURE 2. Outline drawing.

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DEPARTMENT OF THE AIR FORCE

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1071B(EL) ELECTRON TUBE, GAS SWITCHING TYPE 6962

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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